airtel

SYNC

Stadium Surge

Fix the anchor, then let the crowd shield the cells:
a ₹3.52 Cr, data-calibrated plan for a 1,00,000-fan final

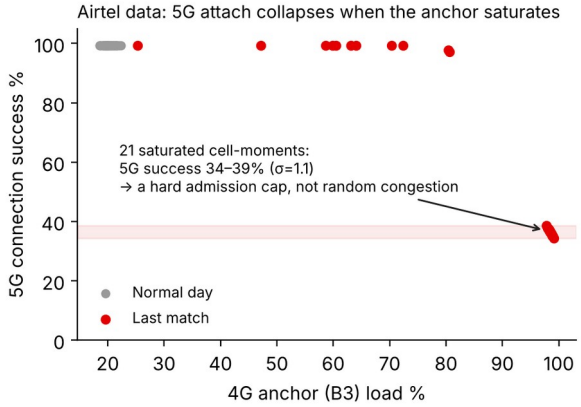
Amritha S & Shreya Kamath
VIT Chennai

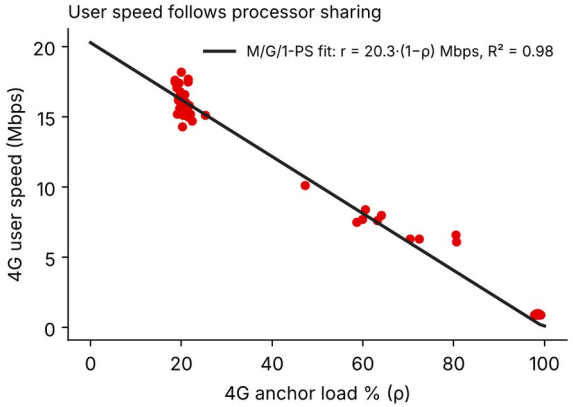


Platform Partner unstop

01 Diagnosis from Airtel's own counters: a hard cap on the 4G anchor

Sync_V1.xlsx, 8 stadium sites × 4 match moments, normal day vs last match (65k fans)





34–39%	5G attach success in all 21 saturated cell-moments (σ = 1.1). Flat = a configured cap, not random congestion
2.7×	attach demand vs anchor capacity at the innings break (from success = C/D)
R² = 0.98	Airtel 4G speeds follow M/G/1 processor sharing: r = 20.3·(1−ρ) Mbps
~24%	5G success at the final if nothing changes (demand × 1.54)

Key insight: 5G capacity was not the bottleneck. NSA 5G needs the B3 anchor to attach and carry its uplink.

- Proof: M6 (anchor 60%) kept 99.2% 5G success. Every cell above 97.8% anchor load sat at about 36%, a flat ceiling that points to an RRC-connected-user / admission limit on B3.
- Stuck 5G phones then add about 2,100 users to the full 4G layer. With processor sharing, speed → 20.3 × (1 − 0.985) ≈ 0.3–1 Mbps, exactly what was measured.
- So the plan fixes the anchor first (signalling) and adds capacity second (spectrum reuse).

02 Task 1: 31.7 Gbps peak needs +15 5G and +27 4G cells

Every cell is planned at 80% load, and traffic is pushed to the layer with the highest spectral efficiency (5G mMIMO)

1,00,000 fans › **40,000** Airtel users (40%) › **28,000 | 12,000** 5G phones | 4G-only › **5,040 | 2,160** online at peak (18%) › **25.2 | 6.5 Gbps** demand @ 5 | 3 Mbps

Cell type (DL)	Spectrum × SE	Capacity	@80%	Users
5G macro, 64T64R mMIMO + MU-MIMO	100 MHz × 74% DL × 16 b/s/Hz	1,184 Mbps	947	189 @5
5G small cell, 4T4R in-bowl	100 MHz × 74% × 7	518 Mbps	414	83 @5
4G macro sector, 3 bands	70 MHz × 2.5	175 Mbps	140	46 @3
4G small cell, 2 bands	40 MHz × 3.0	120 Mbps	96	32 @3

SE assumptions: 4G at 2.5 b/s/Hz cross-checks against the last match (about 3.5k users on 20 cells at about 1 Mbps). The 5G figure assumes MU-MIMO pairing, which works well in a stadium bowl with line of sight.

Requirement @80%	5G	4G
Peak demand	25.2 Gbps	6.5 Gbps
Cells needed (macro-eq.)	27	47
Existing stadium cells*	12	20
Extra cells needed	**+15**	**+27**

*4 macro sites with 5G × 3 sectors; 6 macros × 3 + 2 small cells on 4G

Why the anchor fix comes first

If 5G phones get stuck at last match's 64% failure rate, 4G demand rises from 6.5 to 16.2 Gbps, which means about 69 extra 4G sectors (≈ 23 COWs ≈ ₹5.8 Cr) and breaks the budget. Keeping 5G phones on 5G costs about ₹0 in hardware, and 5G delivers about 6.8× more Mbps per cell, so it is also the most spectrally efficient answer.

03 Task 2: ₹3.52 Cr mix, with 73% spent on assets that stay after the final

Small cells and fibre are permanent; COWs are hired per event. ₹48 L is held in reserve.

Item	Qty	Unit cost	Cost	Adds
Dual-mode small cells (4G+5G), under-seat / handrail, in-bowl	24	₹8 L	₹1.92 Cr	9.9 Gbps 5G + 2.3 Gbps 4G
Cells on Wheels (3-sector 4G + 5G mMIMO), car parks / exits	4	₹25 L	₹1.00 Cr	12 × 5G + 12 × 4G sectors
Fibre backhaul: COWs + small-cell aggregation (2 × 10G)	1.2 km	₹50 L/km	₹0.60 Cr	No backhaul bottleneck
Reserve: feature licences, RRC-user uplift, rehearsal tests	—	—	₹0.48 Cr	10–12% contingency
Total			**₹4.00 Cr**	

Capacity check @80%	5G	4G
Existing cells	11.4	2.8
+ 4 COWs	11.4	1.7
+ 24 small cells	9.9	2.3
Total (Gbps)	32.7	6.8
Peak demand (Gbps)	25.2	6.5
Peak cell load	**62%**	**76%**

Upgrades at no hardware cost

- Enable B1 and B40 as extra NSA anchors
- 256QAM + 4T4R on stadium-facing 4G sectors
- MU-MIMO and 5G uplink on n78 macros
- Move S1/S2 from 1G to 10G on the new fibre ring

Placement logic

- Small cells: stands and concourse (food stalls)
- COWs: 2 at gates and car parks for the exodus, 2 at the stand facing M4/M6, which have no 5G
- Tight tilts and low power to limit interference

Why this mix

- Small cells give the most Mbps per ₹ and stay for every future match
- COWs cover the one-night peak without permanent capex
- Both stay under 80% even with 10% more demand

04 Task 3: keep 5G phones on 5G, and protect payments and emergency calls

Root cause: the NSA B3 anchor ran out of RRC/PRB room, so 5G could not be added and 5G phones fell back to 4G

A. Unstick 5G (the anchor fix)

- Multi-anchor: add B1 and B40 as anchors so B3 is not a single choke point
- Reserve anchor room for 5G-capable phones; steer 4G-only phones to other bands (load balancing, SPID/RFSP)
- Lower the 5G add (B1) threshold and stop releasing 5G when the anchor is congested
- Raise the RRC-connected user licence and shorten inactivity timers during the match
- 5G SA on n78 in the stadium cluster for SA-capable phones, with no anchor needed

→ Target: 5G attach success of 95% or more (was 36%)

B. Protect UPI payments

- UPI uses only kilobytes but cannot wait: give UPI/NPCI and bank endpoints a high-priority QoS rule (PCC + DPI), above video
- Merchant SIMs at stalls get a priority profile (ARP/SPID) and dedicated concourse small cells
- Fallbacks: UPI Lite for small offline payments, UPI 123PAY over voice, and a stall Wi-Fi backup

→ Target: UPI failure under 1.5% (was 7.1%)

C. Guarantee emergency calls

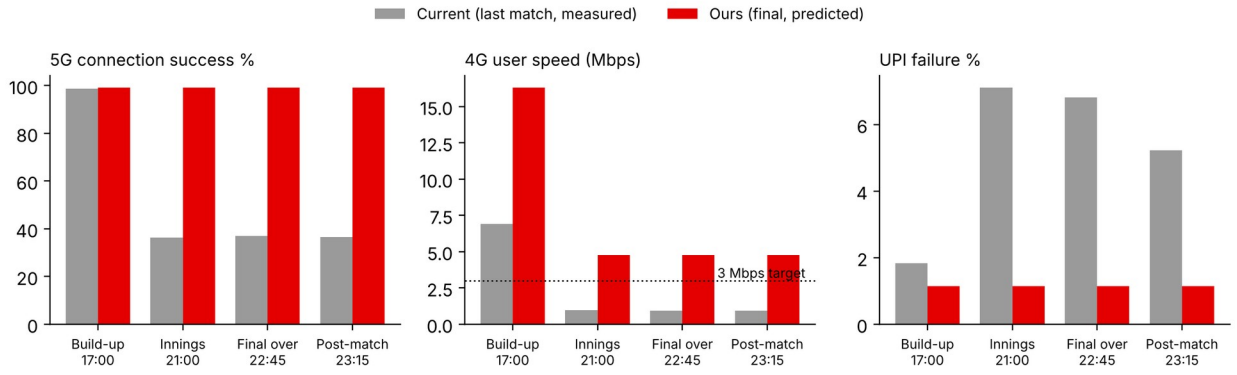
- Emergency (AC10) and VoLTE (QCI 1) are never barred; ARP pre-emption on
- Reserve about 5% of PRBs and RRC slots for QCI 1/5 at every stadium cell
- Priority SIMs (AC 11–15) for police, medical and the stadium ops team
- Access barring switched on automatically for normal users only when load passes 90%

→ Target: 100% emergency-call setup

Order of operations on match night: protect emergency calls first, then UPI payments, then 5G attach, then everything else. The things that matter most can never be pushed out by video.

05 Proof: current vs ours, on Airtel's own data

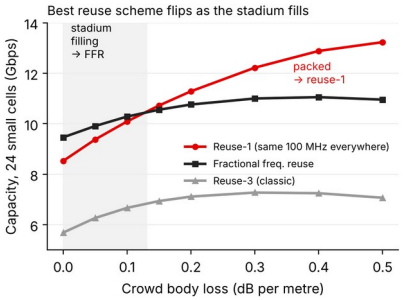
Left: last match (measured) vs final with our plan (predicted, same calibrated models). Bottom: our wireless simulations

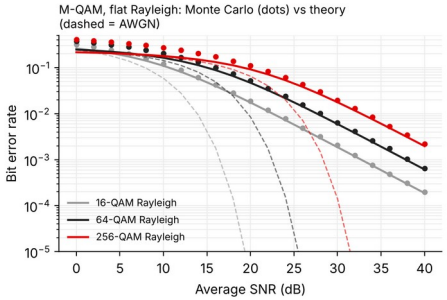


Novelty: the crowd is the shield

Human bodies attenuate 3.5 GHz (3GPP TR 38.901 §7.6.8 blockage). A packed bowl isolates under-seat cells, so the same 100 MHz can be reused in every seat zone.

The network switches FFR → reuse-1 as occupancy rises (ticket scans + ball-by-ball feed).





What the simulations show

- Packed stadium, 24 cells: reuse-1 gives 12.2 Gbps vs 11.0 (FFR) and 7.3 (reuse-3): +66% over classic planning
- Empty stadium: FFR wins, so the scheme must adapt
- Rayleigh M-QAM: Monte Carlo matches theory; 256-QAM needs ~25–30 dB, so SINR gain = more users on 256-QAM
- Python simulation (5 seeds, 5,040 users); MATLAB + Simulink versions included

06 Task 4: yes, there are side-effects. Here is how each one is contained

Most come from packing 40+ new cells into one bowl; all are controlled by design and a live war-room

Possible impact	Why it happens	Mitigation
Interference / lower SINR	Many cells in a small area	Low-power under-seat antennas, downtilt, PCI/PRB planning, tuned for MU-MIMO
Handover ping-pong, dropped calls	Fans move between macro, COW and small cell	Tune hysteresis and time-to-trigger; mobility-robustness SON; test at rehearsal match
Slower video for heavy users	Fair-share and priority rules at peak	Cap video at 720p only above 85% load; nothing below that; prioritise live score and payments
Brief access delay	Access barring for normal users	Only above 90% load, short barring timers, emergency never barred
Neighbourhood coverage changes	Re-tilted macros now face the stadium	Event-only tilt profile, switched on 3 h before and reverted after
Post-match crush at gates and metro	Traffic moves from bowl to exits	2 COWs at exits; capacity shifted by timetable
Extra phone battery drain	NSA dual connectivity (4G + 5G)	SA on n78 removes the dual link; DRX tuning

Live control: an event war-room

A per-cell dashboard (anchor load, 5G attach %, UPI success, VoLTE drops) with automatic triggers at 70/85/90% load. One rehearsal at a league match, 30 days before the final, plus one-click rollback for every parameter change. Triggers are timed to match phases: capacity is boosted 10 minutes before the innings break and the final over.

07 Competitive edge, impact and 60-day roadmap

How mega-events are usually handled, compared with this plan

Approach	Who typically uses it	Gap	This plan
Many COWs everywhere	Indian operators at IPL / big events	Costly every event; interference	4 COWs only, at exits and the weak side
Permanent DAS / in-venue small cells	US operators at Super Bowl venues	High capex, long lead time	24 under-seat cells now, with adaptive reuse
5G SA, no LTE anchor	Jio (SA network in India)	Needs SA core and SA phones	SA in the stadium + NSA multi-anchor
Purpose-built 5G venues	Qatar World Cup 2022 operators	Years of planning	60 days, ₹4 Cr, calibrated on real counters

36% → 99%
5G attach

1 → 4.8 Mbps
4G speed

12 → 70+ Mbps
5G speed

7.1% → ~1.2%
UPI failure

100% setup
Emergency

D-60 Plan & survey

D-45 Fibre + 24 cells

D-30 League-match rehearsal

D-7 COWs + tilt profile

D+1 Keep assets, playbook

Fix the 4G anchor first, then let the crowd shield the cells, so every Airtel fan is on the right layer and every payment and emergency call goes through.

Predictions use models fitted to Sync_V1.xlsx ($R^2 = 0.98$). Crowd-loss value and licence costs to be validated with field measurement and the vendor at the D-30 rehearsal.